3	**(a)**	**(i)**	horizontal velocity = 15 cos 60° = 7.5 m s^{-1}	A1	[1]
		(ii)	vertical velocity = 15 sin 60° = 13 m s^{-1}	A1	[1]
	(b)	**(i)**	$v^2 = u^2 + 2as$ $s = (13)^2 / (2 \times 9.81) = 8.6(1)$ m using g = 10 then max. 1	 A1	 [1]
		(ii)	t = 13 / 9.81 = 1.326 s or t = 9.95 / 7.5 = 1.327 s	A1	[1]
		(iii)	velocity = 6.15 / 1.33 = 4.6 m s^{-1}	M1 A0	 [1]
	(c)	**(i)**	change in momentum = 60×10^{-3} [–4.6 – 7.5] = (–)0.73 N s	C1 A1	 [2]
		(ii)	final velocity / kinetic energy is less after the collision or relative speed of separation < relative speed of approach hence inelastic	 M1 A0	 [1]

4 (a) electrical potential energy (stored) when charge moved and gravitational potential